Week 25 — Extended Reading: Public synthetic audit: evidence and boundaries

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Joachims, Swaminathan and de Rijke (2018): BanditNet

Primary source. https://www.cs.cornell.edu/~tj/publications/joachims_etal_18a.pdf

Mechanism and scope. Deep Learning with Logged Bandit Feedback studies learning from partial feedback with logged propensities. The author-hosted paper is the correct source; arXiv:1707.07875 is an unrelated mathematics paper and must not be used for this citation.

Read, derive, investigate. Explain the difference between a logged randomised display action and a later human decision. Which probability belongs in the weight? A neural policy trained from bandit logs is not automatically a sequential RL system.

2 Geyik, Ambler and Kenthapadi (2019): Fair ranking

Primary source. https://arxiv.org/abs/1905.01989

Mechanism and scope. The paper studies ranking fairness with representation requirements and an application to talent search. Its definition and evaluation population are part of the result; they are not interchangeable with every employment fairness measure.

Read, derive, investigate. Compare a share among the top ten with a group-conditional selection rate among eligible candidates. Use unequal eligible group sizes to show that equal selected counts can still produce different rates. Our constrained toy reports selected shares and should be labelled accordingly.

3 Agrawal and Goyal (2013): Linear Thompson sampling

Primary source. https://arxiv.org/abs/1209.3352

Mechanism and scope. The paper provides a Thompson-sampling analysis under linear-payoff and noise assumptions. A bandit reduction removes action-dependent future transitions; it does not supply these statistical assumptions for an arbitrary recruiting outcome.

Read, derive, investigate. Recompute the four-dimensional posterior example in the lecture. Unobserved coordinates have zero mean but retain prior variance. Separate rank-one update cost from fresh matrix factorisation and candidate scoring cost.

4 Rafailov et al. (2023): DPO in a larger workflow

Primary source. https://arxiv.org/abs/2305.18290

Mechanism and scope. Preference tuning is a possible component of a text-generation workflow. It still needs a defined preference population, supported reference and an independent evaluation of the desired behaviour.

Read, derive, investigate. At policy equals reference, ordinary DPO gives $\log 2$. Does that remain true after adding a nonzero length margin? A preferred outreach message is not necessarily truthful, accessible or effective; list separate tests for those outcomes.

5 Stooke et al. (2020): Constraint feedback

Primary source. https://arxiv.org/abs/2007.03964

Mechanism and scope. Multiplier feedback is useful for understanding the direction of responses to cost violations. It does not guarantee that every parameterisation, stochastic estimate or infeasible constraint converges.

Read, derive, investigate. Explain why an arbitrary fixed penalty does not specify a chosen budget, and why an optimal fixed multiplier can nevertheless recover an ideal constrained optimum. Inspect the actual toy update: it is a heuristic score adjustment, not PPO or a complete PID controller.

6 Official guidance: classification, review and denominators

Primary source. https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

Mechanism and scope. Use official current sources rather than a purported recruitment-specific policy brief. The lecture links the Commission AI Act overview, EDPB-endorsed automated-decision guidance and EEOC selection-procedure guidance, checked 20 October 2026.

Read, derive, investigate. Find the exact intended use, jurisdiction, role, application date and denominator before drawing a conclusion. An optimiser's metric is not a legal determination. No private platform, customer log, or former repository history was accessed for this public synthetic case study.

Connect the reading to your workbook

The central audit experiment is identification, not a rollout timetable. For $k = 10$ and $\varepsilon = 0.05$, derive the exploration floor 0.005 and maximum deterministic-target weight 200. Then construct a target action outside the eligible set: more data from the same logger cannot recover its unobserved outcome without additional assumptions. Keep that distinction visible in every OPE report.
Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

Cross-fitting cannot repair two misspecified nuisance models. Use conditional expectation to reconstruct the DR error product. Record nuisance splits and model assumptions separately; report uncertainty as conditional or full-pipeline, and retain the OPE primary sources for double-robustness conditions.

Checked textbook locations: Siegrist § § 4.7, 7.1, PDF pp.507, 896; Ross § 15.3, PDF p.683. https://stats.libretexts.org/@go/page/10189. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.